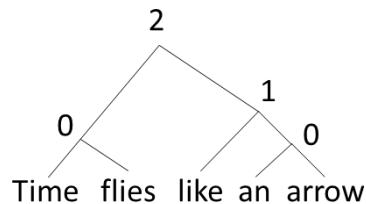


Problem #6: Encoding Trees (20 marks)

Solution:

In the encoding scheme, every internal node is assigned a number, which is equal to its height. In other words, the lowest internal nodes get a label of 0. The next level node gets 1, and so on. A node always gets a label 1 more than the larger of its children's labels. See the diagram below for an example labeling. The number between two successive words x and y is equal to the label of the node which is the lowest common ancestor of x and y .

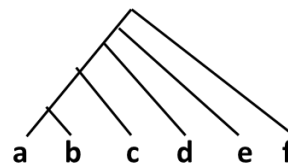
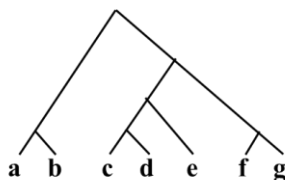


Assignment 1

Tree-C: Time 3 flies 2 like 1 an 0 arrow

Tree-D: Time 0 flies 2 like 1 an 0 arrow

Assignment 2



Assignment 3

- **a 0 b 2 c 0 d 2 e 1 f 0 g** → **a 0 b 3 c 0 d 2 e 1 f 0 g** OR **a 0 b 1 c 0 d 2 e 1 f 0 g**
- **a 1 b 3 c 2 d 1 e 0 f** → **a 0 b 3 c 2 d 1 e 0 f**

Constraints:

In the encoding **a 0 b 2 c 0 d 2 e 1 f 0 g**, the maximum number, which is 2, is present more than once. The constraint it violates is "Every valid string must have one and only one maximum number." This is because the root of the tree, which is the topmost node, is unique and has the highest label.

a 1 b 3 c 2 d 1 e 0 f has no "0" on the left of 3 – the maximum value. Constraint it violates is "If there is at least one number on the left/right of the maximum value, then there must be at least one 0 on the left/right of the maximum value." This is because the maximum value divides the tree into two sub-trees. Any sub-tree, which is also a binary tree, must have at least one 0 in its encoding.

Assignment 4

Upper Bound:

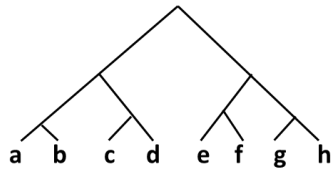
If $s: a_1 x_1 a_2 x_2 a_3 \dots a_{n-1} x_{n-1} a_n$ is a valid tree encoding, where the $\max\{x_i\}$ is the maximum value of all x_i 's, then there must be at least one occurrence of the numbers 0, 1, 2, ... $\max\{x_i\}$ in s , because of the second constraint stated in Assignment (3). Thus, $\max\{x_i\}$ is bounded by $n - 2$. Hence, the maximum value of $\sum_{i=1}^{n-1} 2^{x_i}$ is achieved when $s: a_1 0 a_2 1 a_3 2 \dots a_{n-1} n-2 a_n$.

Therefore,

$$\sum_{i=1}^{n-1} 2^{x_i} \leq 2^0 + 2^1 + 2^2 + \dots + 2^{n-2} = 2^{n-1}$$

Lower Bound:

$\sum_{i=1}^{n-1} 2^{x_i}$ is minimum when $\max\{x_i\}$ is smallest, and the smaller values are repeated more often. This happens when the tree has a balanced structure as shown below.



a 0 b 1 c 0 d 2 e 0 f 1 g 0 h

In such case, at least half of the x_i s are 0, a fourth of the x_i s are 1, an eighth of the x_i s are 2, and so on. Therefore,

$$\begin{aligned} \sum_{i=1}^{n-1} 2^{x_i} &\geq \left\lfloor \frac{n}{2} \right\rfloor 2^0 + \left\lfloor \frac{n}{2^2} \right\rfloor 2^1 + \left\lfloor \frac{n}{2^3} \right\rfloor 2^2 + \dots + \left\lfloor \frac{n}{2^{\log_2 n}} \right\rfloor 2^{\log_2 n-1} \\ &= \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor + \dots + \left\lfloor \frac{n}{2} \right\rfloor \quad (\text{at least } \lfloor \log_2 n \rfloor \text{ times}) \\ &= \lfloor \log_2 n \rfloor \left\lfloor \frac{n}{2} \right\rfloor \end{aligned}$$
